



Lessons Not Learned: We Still Can't Even Get the Basics Right

...OF CS 116 INTRODUCTION TO
SECURITY, SPRING 2026

Before We Begin, Let's
Talk About Dirty
Secrets About This
Class

Dirty Secret 1: Only 25 students read the syllabus

103 students enrolled in CS 116 (Tuesday-Thursday section)

16 students enrolled in CS 116 (Online Masters Wednesday section)

Total: 119 students

21.0% of students read the syllabus (which is an DECREASE from 26.3% in fall 2025)

Emails ranged from January 8th to April 9th

Systems (CS 111), or Computer Architecture (CS 40) yet. Is that a problem?

No. Cyber Security is a very broad field and it is impossible for anyone, even professionals, to know everything. What is important for you is to start thinking about Security.

Q: Will videos be recorded in case I miss class?

If you miss the in-person Tuesday classes due to illness or personal matters, you can always watch the recorded Wednesday sessions for online Master's students (Wednesday sessions are always recorded). Tuesday and Wednesday sessions are practically identical. Thursdays are on Twitch which are recorded and also exported to YouTube.

Q: If I am taking this course for professional purpose, can I have a tuition reimbursement letter or certificate?

A: Absolutely! It's a nice tuition reimbursement letter, hand signed!

If you have read this far, send me an email (ming.chow AT tufts DOT edu) with the subject "Buy and hold" to earn a reward.

Course Policies

Labs

- With the exception of the password cracking lab and CTF writeup, all labs for a given topic, are due on a Sunday at 11:59 PM PDT (that is, Pacific Time).
- With the exception of password cracking lab, the CTF game, and quizzes, you are granted an automatic extension of 24 hours at no cost

Dirty Secret 2: The NSA er... Canvas Puzzle

24 students (or 20% of students) decoded the Base64 string –this is an INCREASE from 8.4% in fall 2025)

“Ard Was Right About Crypto”

Emails ranged from January 11th to April 18th

Puzzle solving and intellectual curiosity are critical skills in Security



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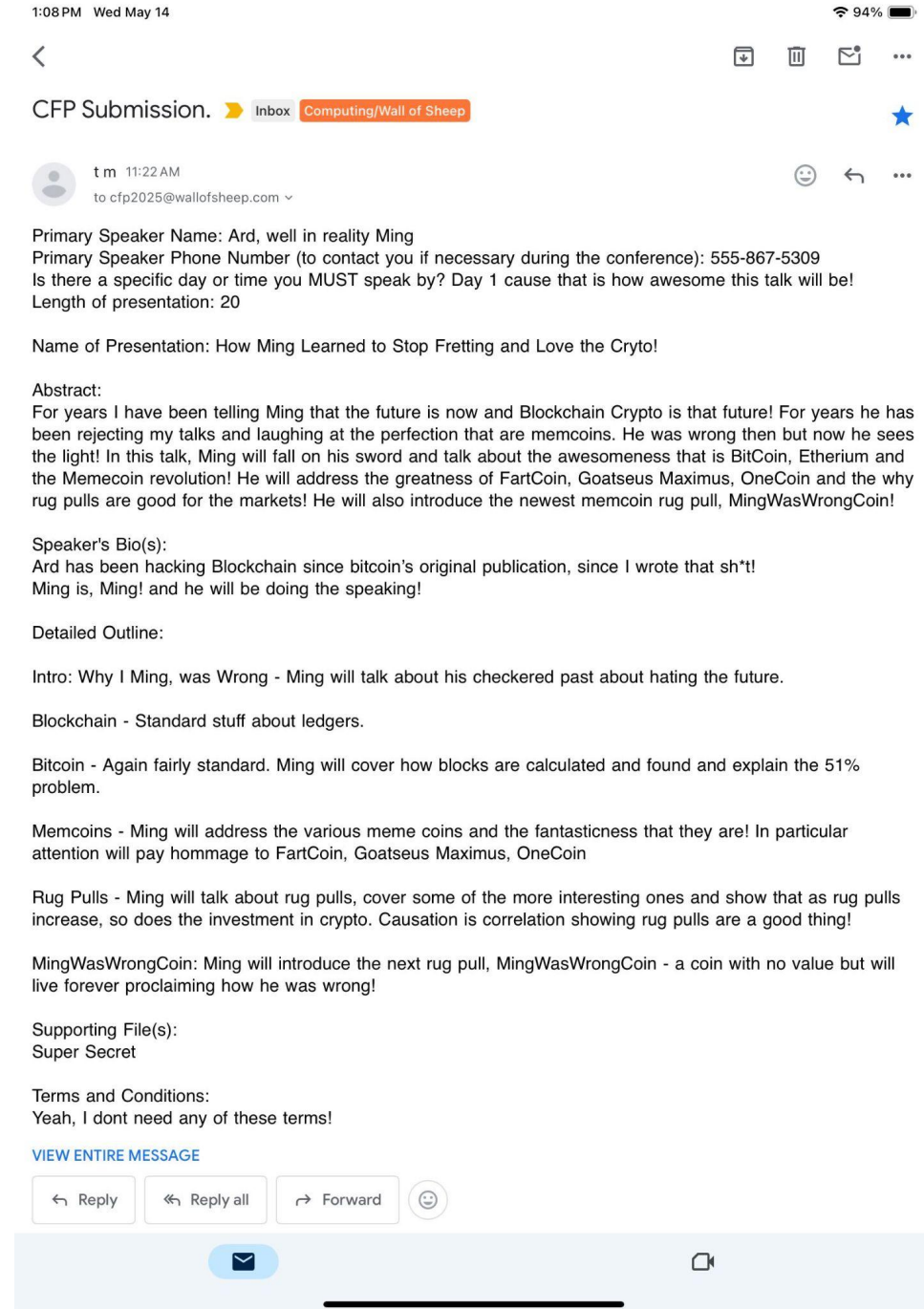
Assign To

Edit

Public Course Website: <https://cs116.org/> . Important: please refer to the course website for most up-to-date syllabus.

Q29uZ3JhdHVzYXRpb25zIGRIY29kaW5nIHRoaXMulE5vdyBlbWFpbCBtZSBhdCBtaW5nLmNob3dAdHVm
dHMuZWR1IHdpdGggdGhliHN1YmplY3QgbGluZTogQXJkIFdhcyBSaWdodCBBYm91dCBDcnlwdG8=

Who Is Ard?



Dirty Secret 3: Special Congrats

...to Marten, Mainak, Yuvit, Emilo, Stephanie, Nick, Edwin, Fahim, Jeffrey, Willa, Ian, Tasnim

...you know what you did!

#LeChampion #JerichoAppreciationSociety #AEW

Dirty Secret 4: About Lab 9, The Technical Risk Analysis Lab

You received either a 5 / 10 or a 10 / 10

The point being: it's not okay to have hardcoded passwords in this day in age. You can't make those mistakes.

Which leads to our feature presentation and underlying message #1 of this course...

And Now For The Feature Presentation

Motivation

- A list of bright shiny objects seen in security products and startups (or buzzword hell)
 - APTs
 - Machine Learning
 - Comprehensive cybersecurity
 - Real-time monitoring
 - Behavioral analysis
 - Next-gen <FILL IN THE BLANKS> (thanks Russell Butturini)
 - Xgen
 - Cloud-enabled

Motivation (continued)

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- ["Top US Undergraduate Computer Science Programs Skip Cybersecurity Classes"](#)
- ["Here are the 61 passwords that powered the Mirai IoT botnet"](#)
- ["One of the Largest Hacks Yet Exposes Data on Hundreds of Thousands of Kids"](#)
- ["Yahoo Says Hackers Stole Data on 500 Million Users in 2014"](#)

Motivation (continued)

Source:

<https://twitter.com/jeremiahg/status/866783974311444480>

 **Jeremiah Grossman** ✓
@jeremiahg

\$81,000,000,000 later: "survey found 35% of companies suffered 2 or more breaches in the last 12mo. 3 in 5 expect to be breached in 2017..."

 **Help Net Security** ✓ @helpnetsecurity · May 18, 2017
3 in 5 companies expect to be breached in 2017 - bit.ly/2rhuwLD



71% aren't sure how to manage and protect unstructured data

6:32 PM · May 22, 2017 · TweetDeck

7 Retweets 3 Quote Tweets 9 Likes

The Gist

- Those motivating slides were from a presentation I gave in 2017
- I could give the same exact talk from 2017 now, without any changes, and get away with it
- In 2013, Veracode gave a presentation “We See the Future and it’s Not Pretty”. The predictions were accurate.
- While a lot of things have changed, a lot have stayed the same...

Key Findings:

- 70% of applications failed to comply with enterprise security policies on first submission.
- SQL injection prevalence has plateaued, affecting approximately 32% of web applications.
- Eradicating SQL injection in web applications remains a challenge as organizations make tradeoffs around what to remediate first.
- Cryptographic issues affect a sizeable portion of Android (64%) and iOS (58%) applications.

Predictions:

- Average CISO Tenure Continues to Decline.
- The Rise of the Everyday Hacker
- Decreased Job Satisfaction/ Higher Turn-over for Security Professionals.
- Default Encryption, Not “Opt-in,” Will Become the Norm.

A Question (Well,
Really Three) For
The Class...

Are We Still Dealing With These Problems?

Phishing and social engineering

Password reuse

Weak passwords

Hardcoded passwords

Distributed Denial of Service (DDoS) attacks?

Malware

Injection

Underlying Message 1: What's the Point in Making This Course Difficult When We Still Can't Even Get The Basics Right?

We've been dealing with the same problems for over 2 decades and counting.

But For Now and For the Future

- Crime will continue to pay --and pay well
- We will still be talking about a Cyber Security skills shortage
 - I've been monitoring how long we've been playing this game for <https://gist.github.com/mchow01/9569350f3b975ce84dad68f0d95c4579>
- There will be another Executive Order on Improving the Nation's (United States) Cybersecurity
- In five years, someone will still be giving a presentation on what SQL injection is, what DDoS is
- In five years, I can give this presentation again without any changes, and get away with it

Underlying Message 2: the Ultimate Goal of This Course: For You To Be A Good Citizen

- We are still battling the same vulnerabilities, having the same debates, for the last 2+ decades
- Your job: talk to others, especially non-tech folks.
 - Example 1: inform developers it is really bad to use C functions like strcpy()
 - Example 2: engage in constructive debates, uncomfortable conversations
- If I still teach this course in 20 years with the same message, then something is terribly wrong
- December 8, 2015: “Congress puts terrorism and tech in the spotlight (CNN)”
<https://money.cnn.com/2015/12/08/technology/encryption-congress-commission/>